

JavaScript
Practical Lab Session 5
Functions & Variable Scope

Name: _ Shailly_____

Roll No.: _2424334_____

Semester: 5

Course: BCA

Part 1 — Function Declaration Basics

Task 1.1 & 1.2

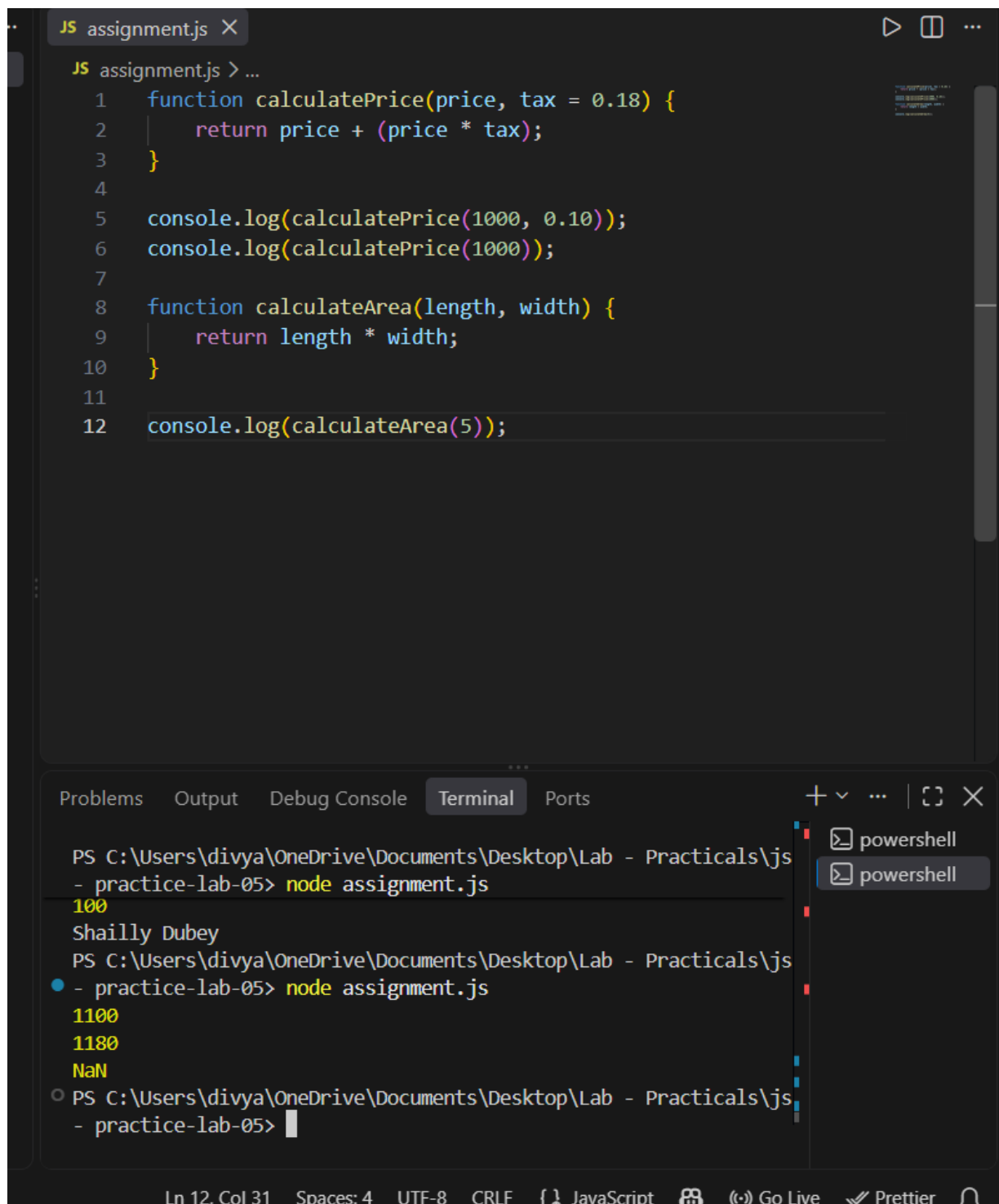
```
JS assignment.js X
JS assignment.js
1 function isAdult(age) {
2   return age >= 18;
3 }
4
5 console.log(isAdult(20));
6 console.log(isAdult(15));
7 console.log(isAdult(18));
8
9 function calculateDiscount(price, isMember) {
10   if (isMember) {
11     return price * 0.9;
12   } else {
13     return price;
14   }
15 }
16
17 console.log(calculateDiscount(1000, true));
18 console.log(calculateDiscount(1000, false));

Problems Output Debug Console Terminal Ports
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js
true
false
true
900
1000
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05>
```

```
JS assignment.js X
JS assignment.js > ...
1  const isAdultExpression = function(age) {
2      return age >= 18;
3  };
4
5  console.log(isAdultExpression(20));
6  console.log(isAdultExpression(16));
7
8  const square = n => n * n;
9
10 console.log(square(2));
11 console.log(square(5));
12 console.log(square(10));
13
14 const fullName = (first, last) => first + " " + last;
15
16 console.log(fullName("Shailly", "Dubey"));

Problems Output Debug Console Terminal Ports
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js>
- practice-lab-05> node assignment.js
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js>
- practice-lab-05> node assignment.js
true
false
4
25
100
Shailly Dubey
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js>
- practice-lab-05>
```

Part 2 — Function Expressions & Arrow Functions



The image shows a Visual Studio Code editor window with a file named `assignment.js` open. The code defines two functions: `calculatePrice` and `calculateArea`. `calculatePrice` takes a `price` and a `tax` (defaulting to 0.18) and returns the price plus tax. `calculateArea` takes a `length` and a `width` and returns their product. The code also includes two `console.log` statements to test these functions.

```
JS assignment.js X
JS assignment.js > ...
1  function calculatePrice(price, tax = 0.18) {
2    |    return price + (price * tax);
3    |  }
4
5    console.log(calculatePrice(1000, 0.10));
6    console.log(calculatePrice(1000));
7
8    function calculateArea(length, width) {
9    |    return length * width;
10   |  }
11
12   console.log(calculateArea(5));
```

The terminal output shows the results of running the code. The first `console.log` statement outputs `1000` and `Shailly Dubey`. The second `console.log` statement outputs `1100`, `1180`, and `NaN`.

```
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js
1000
Shailly Dubey
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js
1100
1180
NaN
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05>
```

The status bar at the bottom indicates the current position is Line 12, Column 31, with 4 spaces, UTF-8 encoding, CRLF line endings, and the file is a JavaScript file. It also shows icons for Go Live and Prettier.

JS assignment.js X

JS assignment.js > ...

```
1 let taxRate = 0.18;
2
3 function finalPrice(amount) {
4   return amount + (amount * taxRate);
5 }
6
7 console.log(finalPrice(1000));
8
9 let taxRate2 = 0.18;
10
11 function testTax() {
12   let taxRate2 = 0.10;
13   console.log("Inside:", taxRate2);
14 }
15
16 testTax();
17 console.log("Outside:", taxRate2);
```

Problems Output Debug Console Terminal Ports

PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js

1180

NaN

PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js

1180

Inside: 0.1

Outside: 0.18

PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05>

powershell

powershell

practice-lab-05

JS assignment.js X

JS assignment.js > ...

```
5
6 console.log(typeof discountApplied);
7
8 if (true) {
9     var discountAppliedVar = true;
10    console.log(discountAppliedVar);
11 }
12
13 console.log(discountAppliedVar);
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js
Outside: 0.18
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js
true
undefined
true
true
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> 
```

Ln 12, Col 1 Spaces: 4 UTF-8 CRLF JavaScript Go Live Prettier

JS assignment.js X

JS assignment.js > ...

```
1  function outerFunction() {
6      }
7
8      innerFunction();
9  }
10
11  outerFunction();
12
13  let role = "guest";
14
15  function loginAsAdmin() {
16      let role = "admin";
17      console.log("Inside function:", role);
18  }
19
20  loginAsAdmin();
21  console.log("Global role:", role);
```

Problems

Output

Debug Console

Terminal

Ports

PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js - practice-lab-05> node assignment.js

true

true

PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js - practice-lab-05> node assignment.js

Hello from outer function

Inside function: admin

Global role: guest

PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js - practice-lab-05>

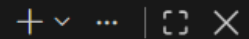
JS assignment.js X



JS assignment.js > ...

```
1  feeCollected = 0;
2
3  calculateGrade(marks) {
4    marks >= 90) {
5    return "A";
6    if (marks >= 75) {
7    return "B";
8    if (marks >= 60) {
9    return "C";
10   {
11   return "F";
12
13
14
15   calculateLateFee = function(daysLate = 0) {
16     daysLate * 10;
17
18
19   processStudent = (name, marks, daysLate = 0) => {
20     grade = calculateGrade(marks);
21     lateFee = calculateLateFee(daysLate);
22
23     feeCollected += lateFee;
```

Problems Output Debug Console Terminal Ports



```
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js
Node.js v24.19.0
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js
Aditi → Grade A, Late Fee Rs.0
Rohit → Grade C, Late Fee Rs.30
Meera → Grade F, Late Fee Rs.50
Final totalFeeCollected → Rs.80
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> 
```

powershell

powershell

Ln 34, Col 66 Spaces: 4 UTF-8 CRLF JavaScript Go Live Prettier

JS assignment.js X

JS assignment.js > ...

```
14
15
16  alance = 1000;
17
18  ion withdraw(amount) {
19    alance = balance - amount;
20    eturn balance;
21
22
23    raw(200);
24    le.log(balance);
25
26
27    sayHello = function() {
28      onsole.log("Hi!");
29
30
31    llo();
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js
Final totalFeeCollected → Rs.80
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> node assignment.js
8
20
800
Hi!
PS C:\Users\divya\OneDrive\Documents\Desktop\Lab - Practicals\js
- practice-lab-05> 
```

+ v ... | [] X
powershell
powershell

Navigate to Command
Scroll to Previous Command (Ctrl+UpArrow)
Scroll to Next Command (Ctrl+DownArrow)

Ln 31, Col 12 Spaces: 4 UTF-8 CRLF { } JavaScript Go Live Prettier